

Digital Management System for Persatuan Kenyah Badeng Sarawak (KEBANA)

Empowering Sarawak's Rural Communities through Secure, Unified, and Centralized Operational Infrastructure.

PROJECT OVERVIEW

Rural community associations, like **Persatuan Kenyah Badeng Sarawak (KEBANA)**, face persistent digital divides. Currently reliant on manual paper archives and decentralized spreadsheets, operational workflows suffer from major coordination gaps.

This project introduces the **KEBANA Digital Management System (KDMS)**, an internal backend platform built to serve as a **Single Source of Truth (SSoT)**. By digitizing administrative tasks, KDMS bridges the accessibility gap, supporting Sarawak's PCDS 2030 digital economy initiatives.

PROBLEM STATEMENT

Geographical Dispersion of Leaders

Committee members are spread across cities and remote Sarawak settlements, impeding synchronized physical data handovers.

Data Incoherency & Loss

Scattered local Excel sheets and WhatsApp logs lead to record duplication, file corruption, and loss of institutional history.

Administrative Overhead

Manual calculations of community finances, membership tallies, and proposal tracking consume excessive administrative hours.

PROJECT OBJECTIVES

1

Centralize Records

Develop a unified database for all KEBANA members with duplicate prevention safeguards.

2

Digital Archiving

Implement a file-hosting flow to create event proposals, store minutes, and track approvals.

3

Automate Ledger

Design a digital cash book that calculates balances and auto-generates monthly summaries.

SYSTEM ARCHITECTURE

KDMS uses a lightweight **Model-View-Template (MVT)** architecture optimized for lower internet bandwidth.

Client View

Bootstrap 5 / JS



Backend Core

Python / Django



Database

MySQL (Relational)

TECHNOLOGY STACK

BACKEND INFRASTRUCTURE

Python Django Framework

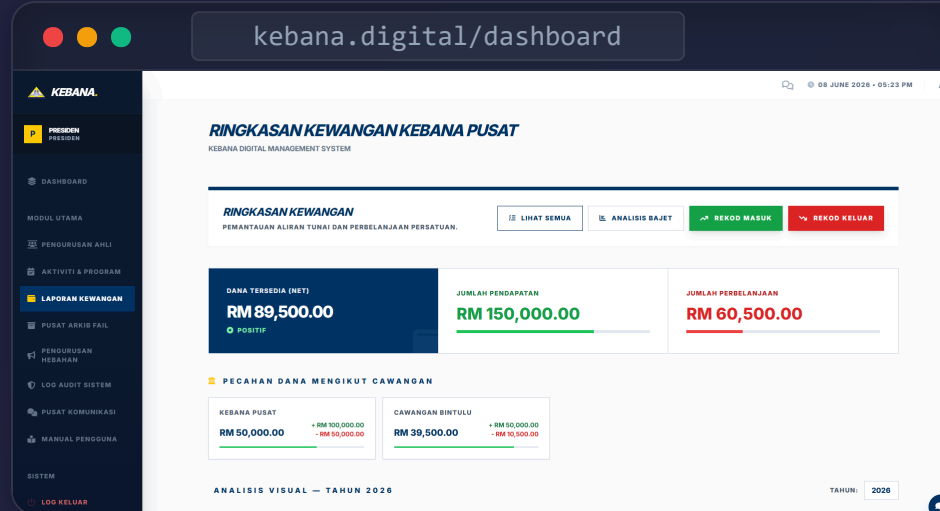
FRONTEND & UI

Bootstrap 5 HTML5 CSS3 JavaScript

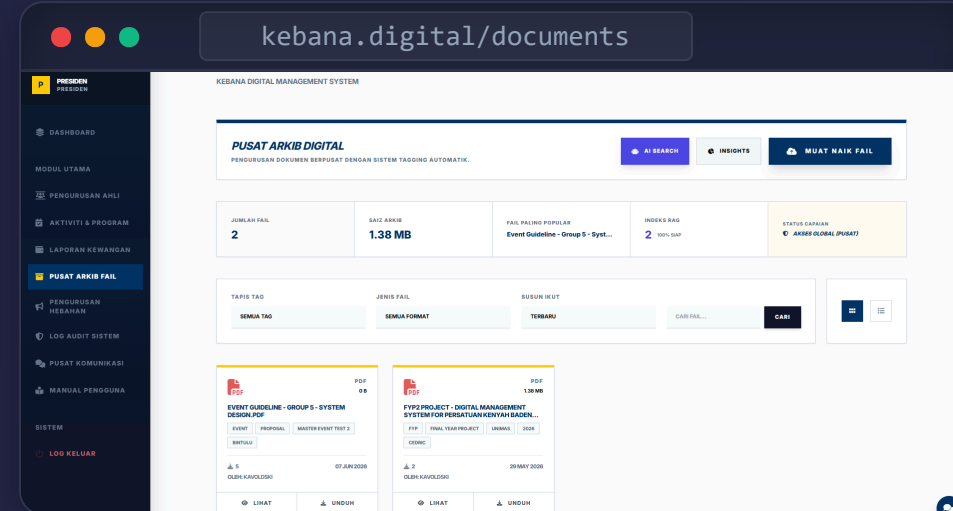
DATABASE & VERSION CONTROL

MySQL Database Git Versioning

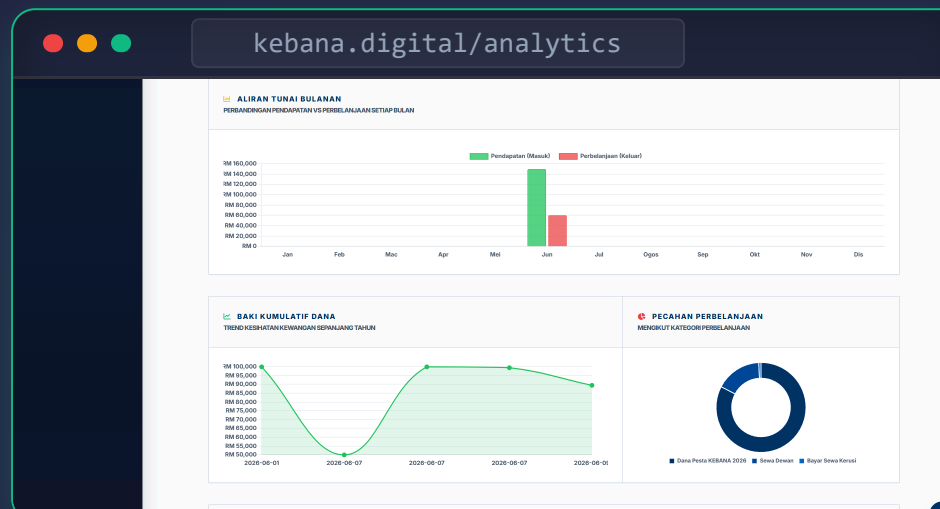
SYSTEM INTERFACE SHOWCASE



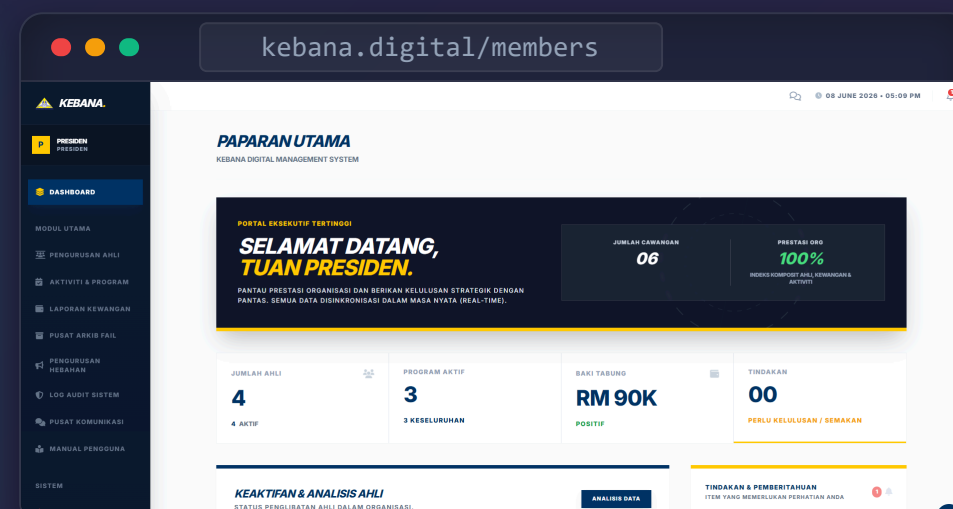
Financial Dashboard: Automated financial ledger providing real-time transparency.



Document Hub: Centralized document hub for secure archiving and proposal management.



Analytics Hub: Data analytics tracking community funds and historical trends.



Member Registry: Searchable records and profile management.

CORE SYSTEM MODULES



Membership Registry

Enables full profile CRUD. Implements IC validations, searchable tables, and automated total counter dashboard widgets.



Event Document Hub

A digital repository where committee members draft proposals, upload letters, and archive community project files.



Financial Cash Book

Automated tracker replacing manual ledger math. Displays itemized income/expense entries and outputs PDF audits.



Role-Based Access

Ensures security integrity. Access levels: President (Superuser), Secretary (Events & Members), Treasurer (Finances).



METHODOLOGY

The system was developed under the **Agile Software Development Lifecycle**, utilizing iterative feedback loops.

1

Gather

2

Design

3

Sprint

4

UAT

5

Deploy



SIGNIFICANCE & OUTCOMES

- 90% Time Savings:** Instant digital lookups replace manual file browsing.
- Institutional Archival:** Protects official community records against fire or resignation loss.
- Auditable Ledger:** Enhances community fund transparency through automated calculation logic.



FUTURE ENHANCEMENTS

- Mobile App Integration**
Provide offline capability and push notifications for news, emergency updates, and meeting invitations.
- Online Payment Gateway**
Automate membership fee collection and donation tracking through integration with local payment solutions (e.g., Sarawak Pay).
- Data Analytics & Reporting**
Enable demographic data visualization and interactive financial trends for the executive committee.



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